

Chapter

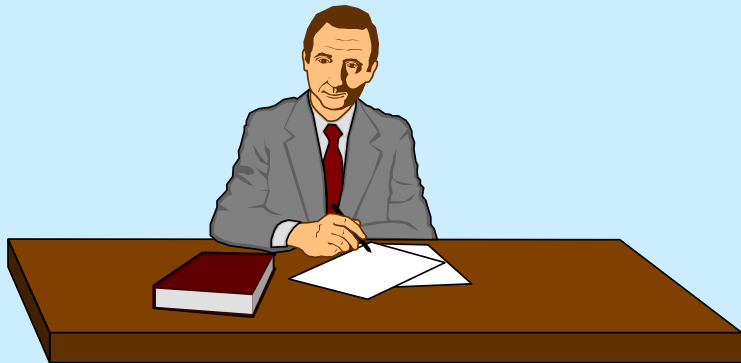
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Introduction to Managerial Accounting and Cost Concepts

Managerial Accounting and Financial Accounting

Managerial accounting provides information for managers of an organization who direct and control its operations.



Financial accounting provides information to stockholders, creditors and others who are outside the organization.



Work of Management

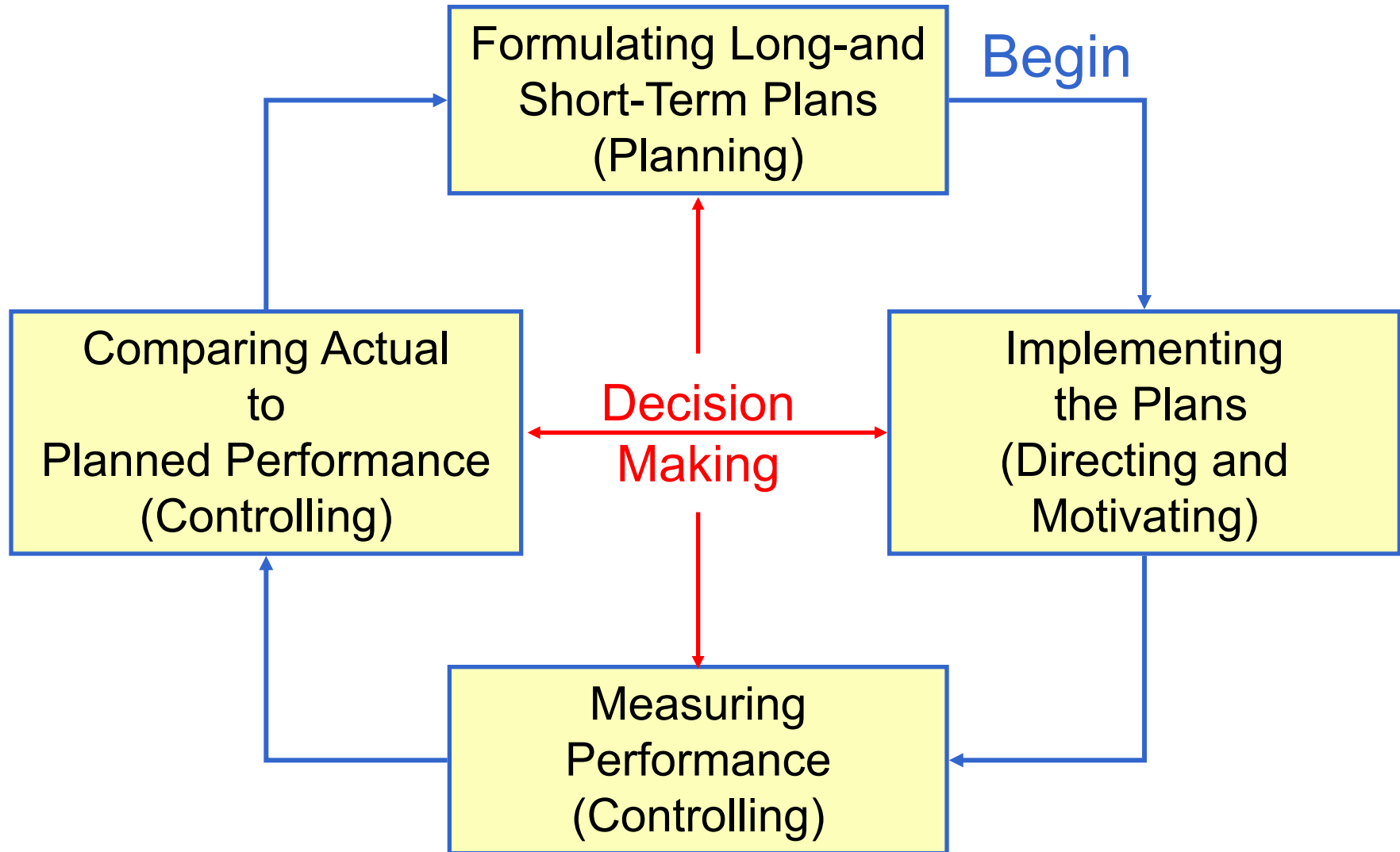
Planning

**Directing and
Motivating**

Controlling



Planning and Control Cycle



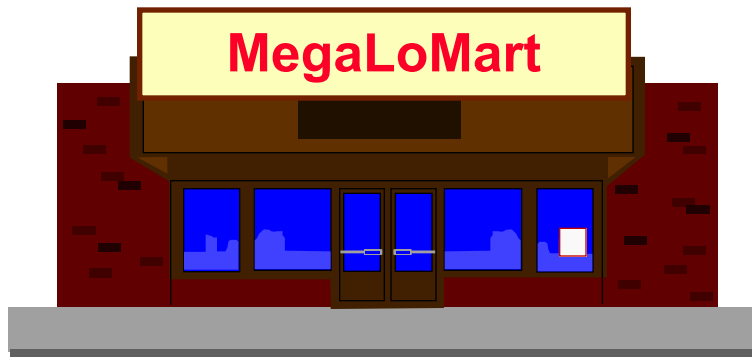
Differences Between Financial and Managerial Accounting

	Financial Accounting	Managerial Accounting
1. Users	External persons who make financial decisions	Managers who plan for and control an organization
2. Time focus	Historical perspective	Future emphasis
3. Verifiability versus relevance	Emphasis on verifiability	Emphasis on relevance for planning and control
4. Precision versus timeliness	Emphasis on precision	Emphasis on timeliness
5. Subject	Primary focus is on the whole organization	Focuses on segments of an organization
6. Requirements	Must follow GAAP and prescribed formats	Need not follow GAAP or any prescribed format

Comparing Merchandising and Manufacturing Activities

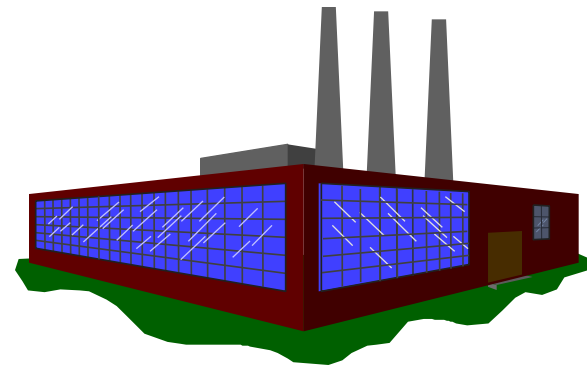
Merchandisers . . .

- ❖ Buy finished goods.
- ❖ Sell finished goods.

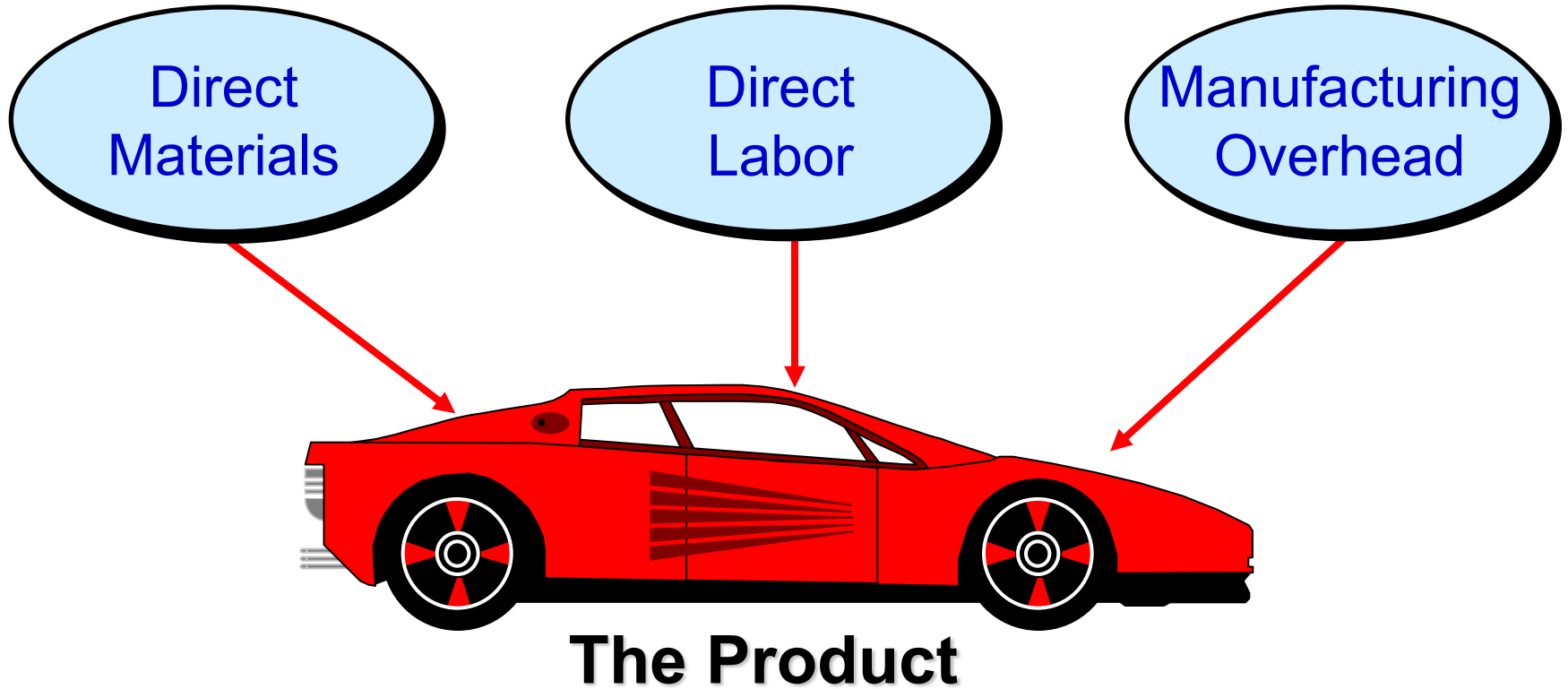


Manufacturers . . .

- ❖ Buy raw materials.
- ❖ Produce and sell finished goods.

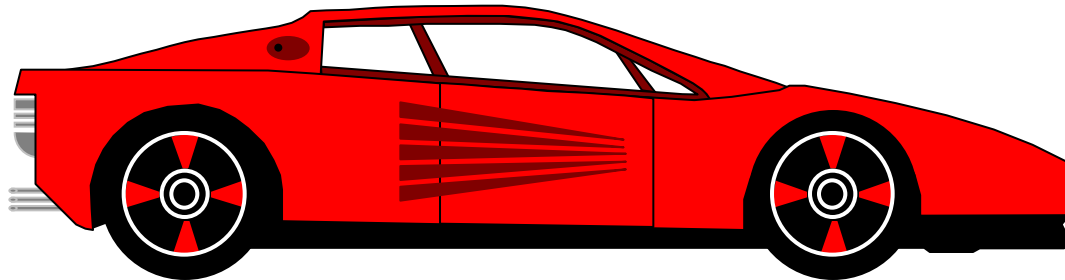


Manufacturing Costs



Direct Materials

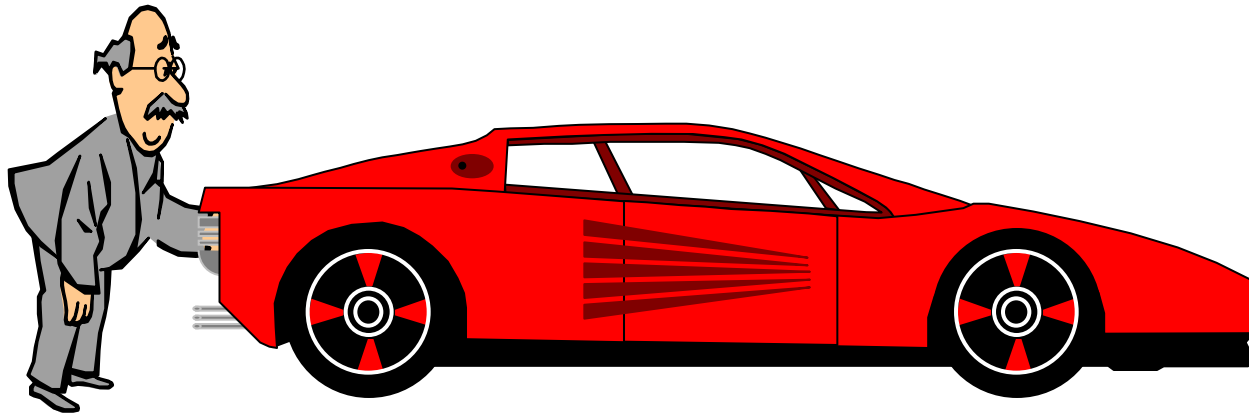
Those materials that become an integral part of the product and that can be conveniently traced directly to it.



Example: A radio installed in an automobile

Direct Labor

Those labor costs that can be easily traced to individual units of product.



Example: Wages paid to automobile assembly workers

Manufacturing Overhead

Manufacturing costs that **cannot** be traced directly to specific units produced.

Examples: Indirect labor and indirect materials

Wages paid to employees who are not directly involved in production work.

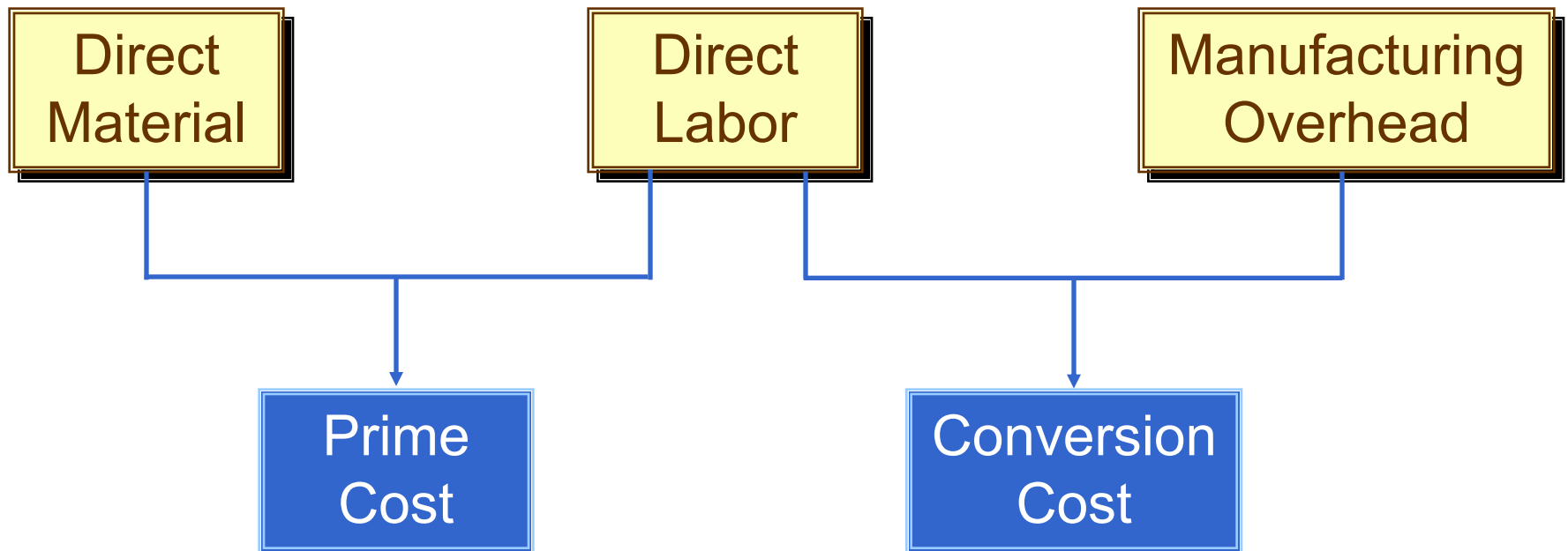
Examples: maintenance workers, janitors and security guards.

Materials used to support the production process.

Examples: lubricants and cleaning supplies used in the automobile assembly plant.

Classifications of Costs

Manufacturing costs are often classified as follows:



Nonmanufacturing Costs

Marketing and selling costs . . .

- ❖ Costs necessary to get the order and deliver the product.

Administrative costs . . .

- ❖ All executive, organizational, and clerical costs.



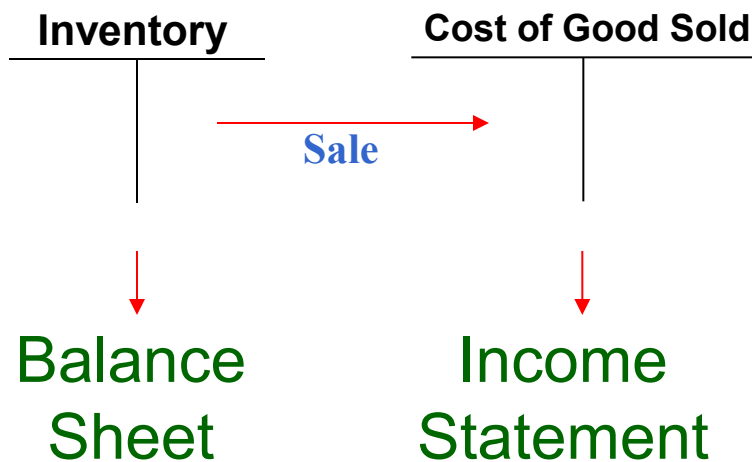
Quick Check ✓

Which of the following costs would be considered manufacturing overhead at Boeing? (More than one answer may be correct.)

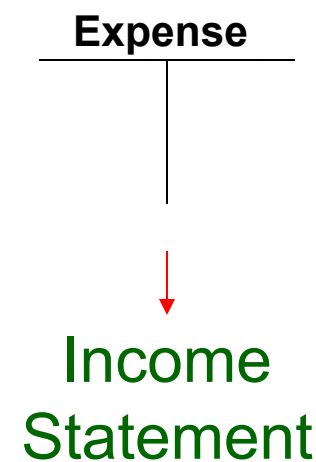
- A. Depreciation on factory forklift trucks.
- B. Sales commissions.
- C. The cost of a flight recorder in a Boeing 767.
- D. The wages of a production shift supervisor.

Product Costs Versus Period Costs

Product costs include direct materials, direct labor, and manufacturing overhead.



Period costs are not included in product costs. They are expensed on the income statement.



Quick Check ✓

Which of the following costs would be considered a period rather than a product cost in a manufacturing company?

- A. Manufacturing equipment depreciation.
- B. Property taxes on corporate headquarters.
- C. Direct materials costs.
- D. Electrical costs to light the production facility.

Balance Sheet

Merchandiser

Current assets

- ❖ Cash
- ❖ Receivables
- ❖ Prepaid expenses
- ❖ Merchandise inventory

Manufacturer

Current Assets

- Cash
- Receivables
- Prepaid Expenses
- Inventories

Raw Materials

Work in Process

Finished Goods

Balance Sheet

Merchandiser

Current assets

- ❖ Cash
- ❖ Receivables
- ❖ Prepaid expenses

Partially complete products – some material, labor, or overhead has been added.

Manufacturer

Current Assets

- Cash

Materials waiting to be processed.

- Inventories

Raw Materials

Work in Process

Finished Goods

Completed products awaiting sale.



The Income Statement

Cost of goods sold for manufacturers differs only slightly from cost of goods sold for merchandisers.

Merchandising Company

Cost of goods sold:

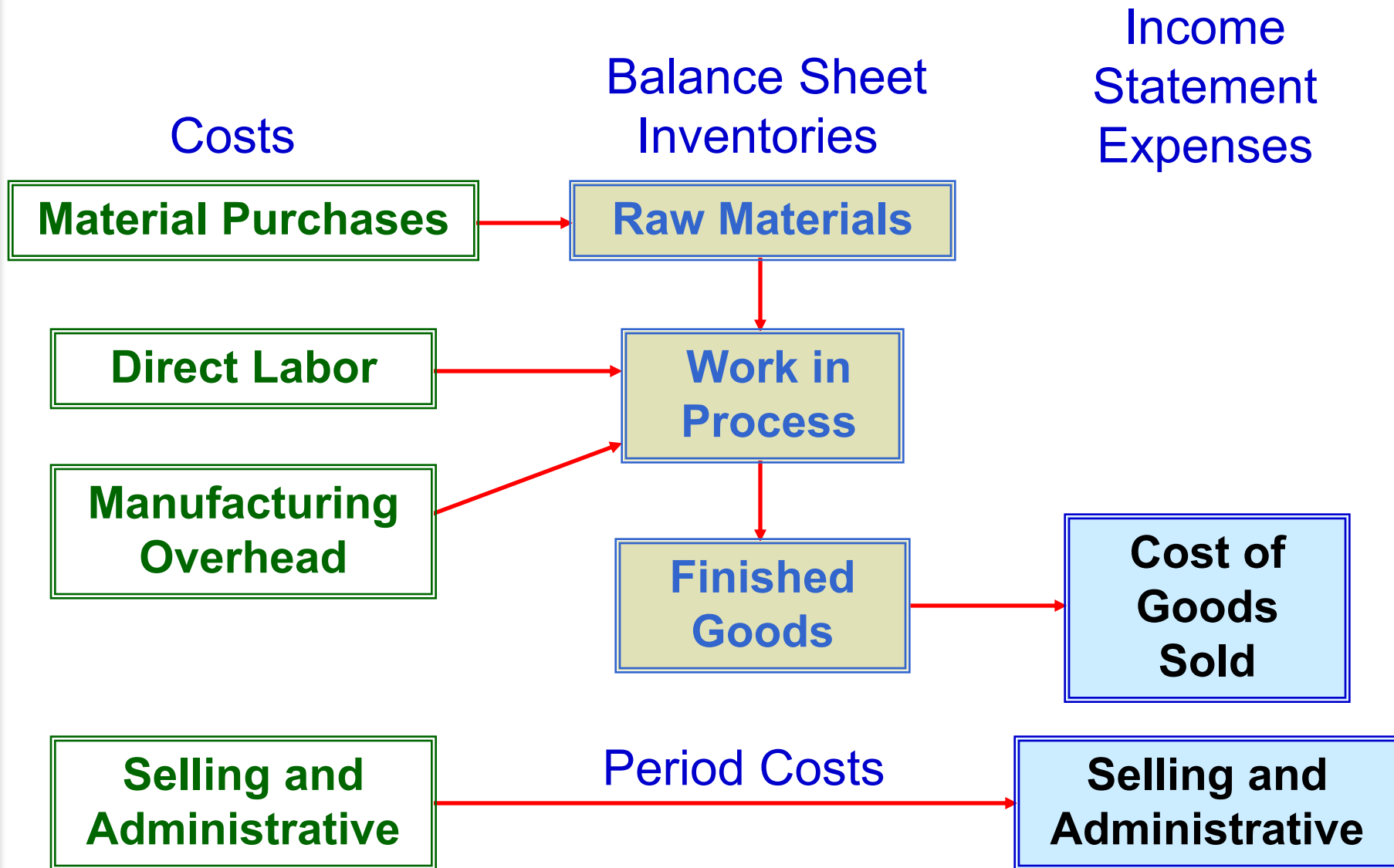
Beg. merchandise inventory	\$ 14,200
+ Purchases	<u>234,150</u>
Goods available for sale	\$ 248,350
- Ending merchandise inventory	<u>(12,100)</u>
= Cost of goods sold	<u><u>\$ 236,250</u></u>

Manufacturing Company

Cost of goods sold:

Beg. finished goods inv.	\$ 14,200
+ Cost of goods manufactured	<u>234,150</u>
Goods available for sale	\$248,350
- Ending finished goods inventory	<u>(12,100)</u>
= Cost of goods sold	<u><u>\$236,250</u></u>

Manufacturing Cost Flows



Quick Check ✓

Which of the following transactions would immediately result in an expense? (There may be more than one correct answer.)

- A. Work in process is completed.
- B. Finished goods are sold.
- C. Raw materials are placed into production.
- D. Administrative salaries are accrued and paid.

Inventory Flows

$$\begin{array}{|c|} \hline \text{Beginning} \\ \text{balance} \\ \$\$ \end{array} + \begin{array}{|c|} \hline \text{Additions} \\ \$\$\$ \end{array} = \begin{array}{|c|} \hline \text{Available} \\ \$\$\$\$\$ \end{array}$$

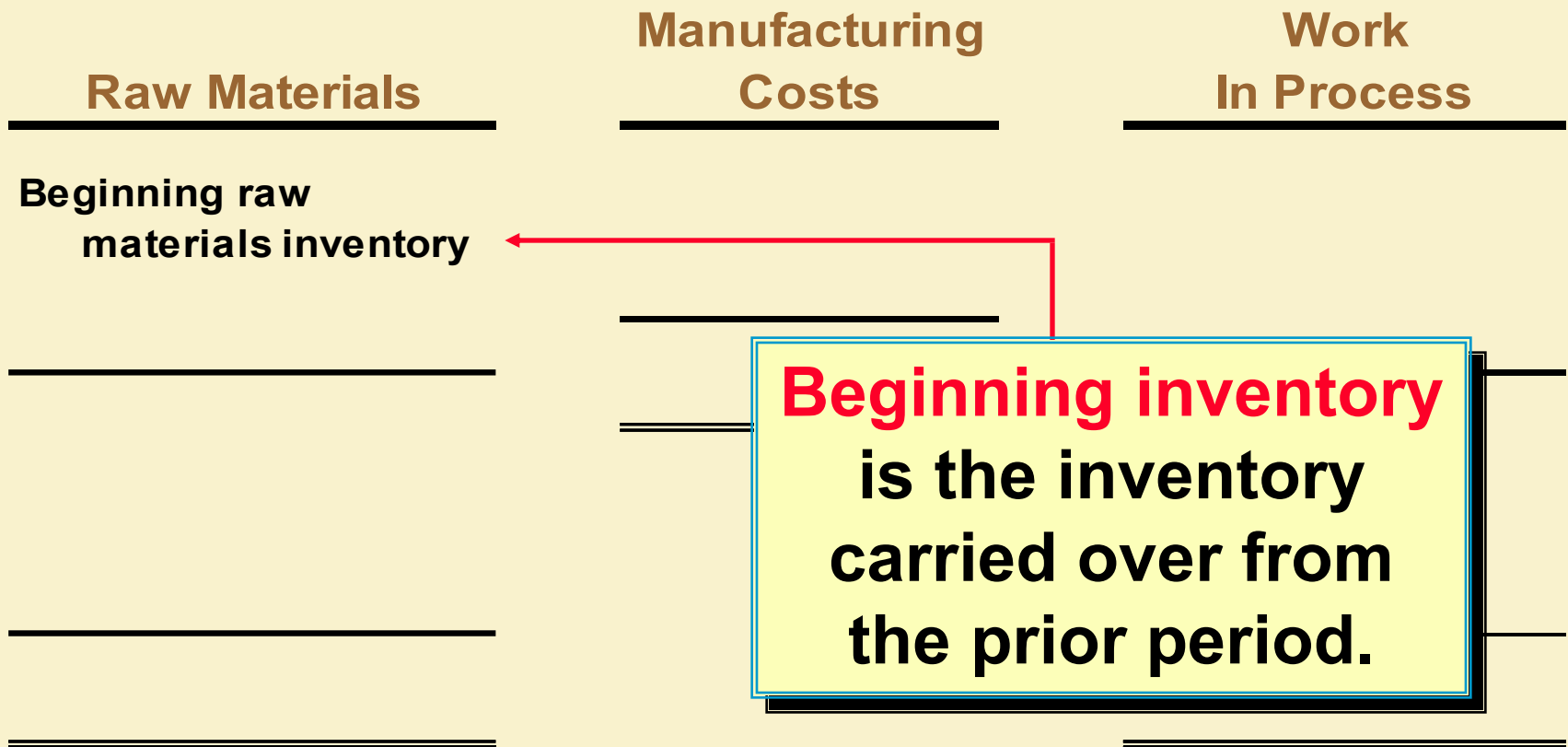
$$\begin{array}{|c|} \hline \text{Available} \\ \$\$\$\$\$ \end{array} - \begin{array}{|c|} \hline \text{Withdrawals} \\ \$\$\$ \end{array} = \begin{array}{|c|} \hline \text{Ending} \\ \text{balance} \\ \$\$ \end{array}$$

Quick Check ✓

If your bank balance at the beginning of the month was \$1,000, you deposited \$100 during the month, and withdrew \$300 during the month, what would be the balance at the end of the month?

- A. \$1,000.
- B. \$ 800.
- C. \$1,200.
- D. \$ 200.

Product Costs - A Closer Look



Product Costs - A Closer Look

Raw Materials

Beginning raw materials inventory
 + Raw materials purchased
 = Raw materials available for use in production
 - Ending raw materials inventory
 = **Raw materials used in production**

Manufacturing Costs

Direct materials

Work In Process

As items are removed from raw materials inventory and placed into the production process, they are called direct materials.

Quick Check ✓

Beginning raw materials inventory was \$32,000. During the month, \$276,000 of raw material was purchased. A count at the end of the month revealed that \$28,000 of raw material was still present. What is the cost of direct material used?

- A. \$276,000
- B. \$272,000
- C. \$280,000
- D. \$ 2,000

Product Costs - A Closer Look

Raw Materials	Manufacturing Costs	Work In Process
Beginning raw materials inventory	Direct materials	
+ Raw materials purchased	+ Direct labor	
	+ Mfg. overhead	
= Raw materials available for use in production	= Total manufacturing costs	
- Ending raw materials inventory		
= Raw materials used in production		

Product Costs - A Closer Look

Raw Materials

Beginning raw materials inventory
+ Raw materials purchased
= Raw materials available for use in production
– Ending raw materials inventory
= Raw materials used in production

Manufacturing Costs

Direct materials
+ Direct labor
+ Mfg. overhead
= Total manufacturing costs

Work In Process

Conversion costs are costs incurred to convert the direct material into a finished product.



Quick Check ✓

Direct materials used in production totaled \$280,000. Direct labor was \$375,000 and factory overhead was \$180,000. What were total manufacturing costs incurred for the month?

- A. \$555,000
- B. \$835,000
- C. \$655,000
- D. Cannot be determined.

Product Costs - A Closer Look

<u>Raw Materials</u>	<u>Manufacturing Costs</u>	<u>Work In Process</u>
Beginning raw materials inventory	Direct materials	Beginning work in process inventory
+ Raw materials purchased	+ Direct labor	
	+ <u>Mfg. overhead</u>	+ Total manufacturing costs
= Raw materials available for use in production	= <u>Total manufacturing costs</u>	= Total work in process for the period
- Ending raw materials inventory		
= <u>Raw materials used in production</u>		

All manufacturing costs incurred during the period are added to the beginning balance of work in process.

Product Costs - A Closer Look

Raw Materials

Beginning raw materials inventory
+ Raw materials purchased

= Raw materials available for use in production

Manufacturing Costs

Direct materials
+ Direct labor
+ Mfg. overhead
= Total manufacturing costs

Work In Process

Beginning work in process inventory
+ Total manufacturing costs

= Total work in process for the period
– Ending work in process inventory

= **Cost of goods manufactured.**

Costs associated with the goods that are completed during the period are transferred to finished goods inventory.



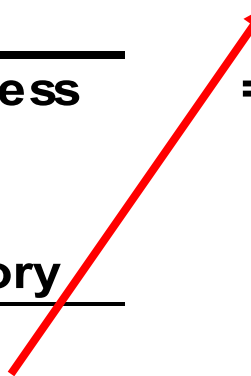
Quick Check ✓

Beginning work in process was \$125,000. Manufacturing costs incurred for the month were \$835,000. There were \$200,000 of partially finished goods remaining in work in process inventory at the end of the month. What was the cost of goods manufactured during the month?

- A. \$1,160,000
- B. \$ 910,000
- C. \$ 760,000
- D. Cannot be determined.

Product Costs - A Closer Look

Work In Process	Finished Goods
Beginning work in process inventory	Beginning finished goods inventory
+ Manufacturing costs for the period	+ Cost of goods manufactured
= Total work in process for the period	= Cost of goods available for sale
- Ending work in process inventory	- Ending finished goods inventory
= Cost of goods manufactured	Cost of goods sold



Quick Check ✓

Beginning finished goods inventory was \$130,000. The cost of goods manufactured for the month was \$760,000. And the ending finished goods inventory was \$150,000. What was the cost of goods sold for the month?

- A. \$ 20,000.
- B. \$740,000.
- C. \$780,000.
- D. \$760,000.

Cost Classifications for Predicting Cost Behavior

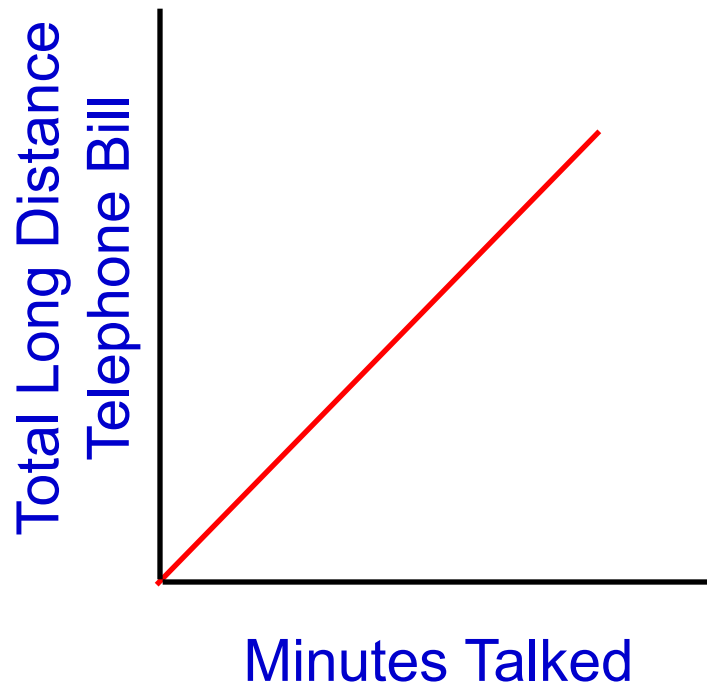


How a cost will react to changes in the level of business activity.

- ❖ Total **variable costs** change when activity changes.
- ❖ Total **fixed costs** remain unchanged when activity changes.

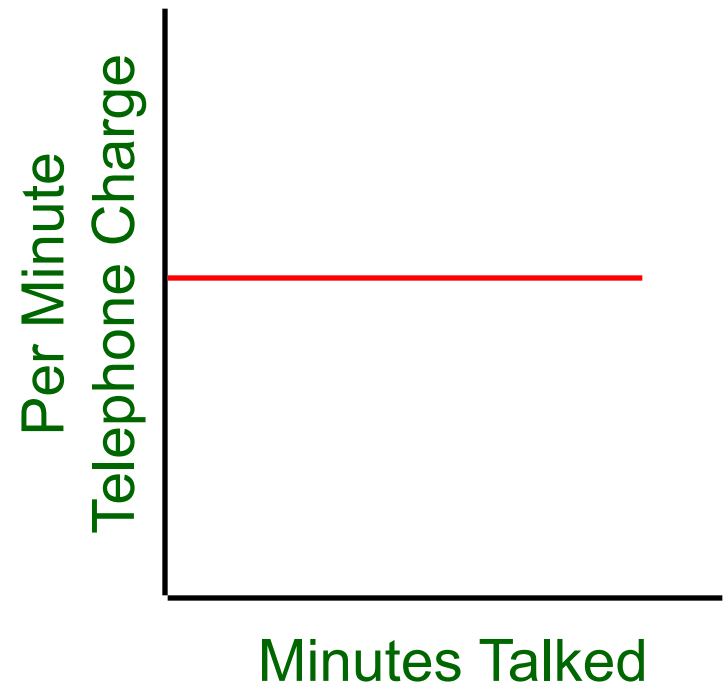
Total Variable Cost

Your **total long distance** telephone bill is based on how many minutes you talk.



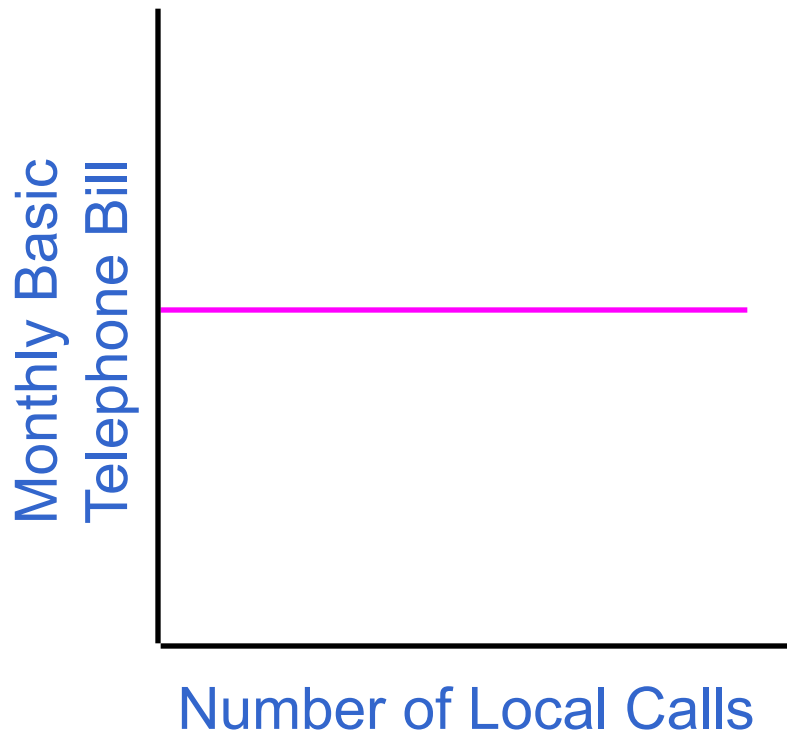
Variable Cost Per Unit

The **cost per long distance minute** talked is constant. For example, 10 cents per minute.



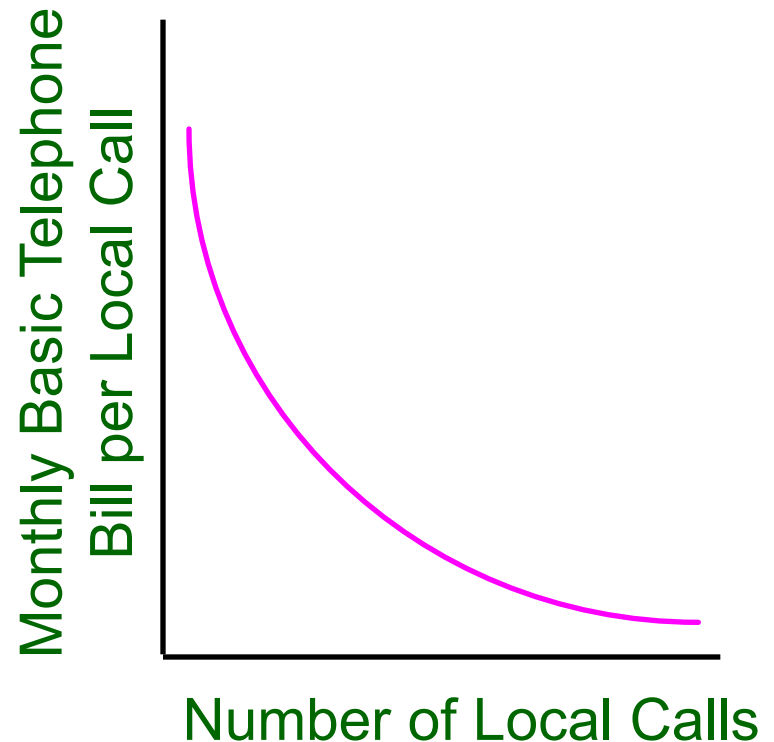
Total Fixed Cost

Your monthly **basic telephone bill** probably does not change when you make more local calls.



Fixed Cost Per Unit

The average cost **per local call** decreases as more local calls are made.



Cost Classifications for Predicting Cost Behavior

Behavior of Cost (within the relevant range)		
Cost	In Total	Per Unit
Variable	Total variable cost changes as activity level changes.	Variable cost per unit remains the same over wide ranges of activity.
Fixed	Total fixed cost remains the same even when the activity level changes.	Fixed cost per unit goes down as activity level goes up.

Quick Check ✓

Which of the following costs would be variable with respect to the number of cones sold at a Baskins & Robbins shop? (There may be more than one correct answer.)

- A. The cost of lighting the store.
- B. The wages of the store manager.
- C. The cost of ice cream.
- D. The cost of napkins for customers.

Quick Check ✓

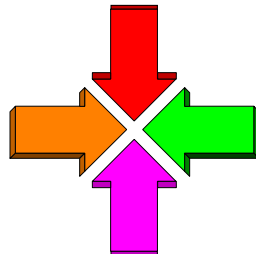
Which of the following costs would be variable with respect to the number of people who buy a ticket for a show at a movie theater? (There may be more than one correct answer.)

- A. The cost of renting the film.
- B. Royalties on ticket sales.
- C. Wage and salary costs of theater employees.
- D. The cost of cleaning up after the show.

Direct Costs and Indirect Costs

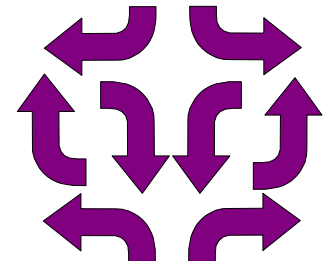
Direct costs

- Costs that can be easily and conveniently traced to a unit of product or other cost objective.
- Examples: direct material and direct labor



Indirect costs

- Costs cannot be easily and conveniently traced to a unit of product or other cost object.
- Example: manufacturing overhead



Differential Costs and Revenues

Costs and revenues that differ among alternatives.

Example: You have a job paying \$1,500 per month in your hometown. You have a job offer in a neighboring city that pays \$2,000 per month. The commuting cost to the city is \$300 per month.

Differential revenue is:

$$\text{\$2,000} - \text{\$1,500} = \text{\$500}$$

Differential cost is:

\$300

Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the cost of the pizza you ate last night relevant in this decision? In other words, should the cost of the pizza affect the decision of whether you drive or take the train to Portland?

- A. Yes, the cost of the pizza is relevant.
- B. No, the cost of the pizza is not relevant.

Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the cost of the train ticket relevant in this decision? In other words, should the cost of the train ticket affect the decision of whether you drive or take the train to Portland?

- A. Yes, the cost of the train ticket is relevant.
- B. No, the cost of the train ticket is not relevant.

Note



- Every decision involves a choice from among at least two alternatives.
- Only those costs and benefits that differ between alternatives (i.E., Differential costs and benefits) are relevant in a decision. All other costs and benefits can and should be ignored.

Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the annual cost of licensing your car relevant in this decision?

- A. Yes, the licensing cost is relevant.
- B. No, the licensing cost is not relevant.

Quick Check ✓

Suppose you are trying to decide whether to drive or take the train to Portland to attend a concert. You have ample cash to do either, but you don't want to waste money needlessly. Is the depreciation on your car relevant in this decision?

- A. Yes, the depreciation is relevant.
- B. No, the depreciation is not relevant.

Opportunity Costs

The potential benefit that is given up when one alternative is selected over another.

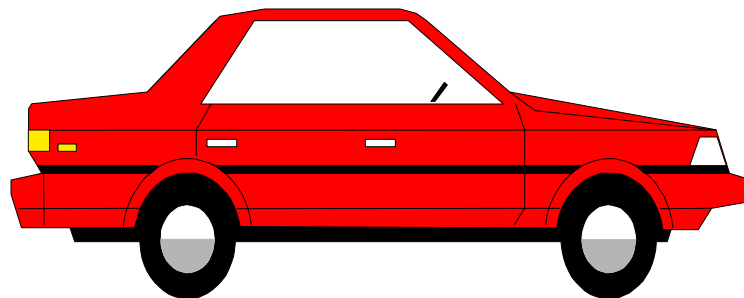
Example: If you were not attending college, you could be earning \$15,000 per year. Your opportunity cost of attending college for one year is \$15,000.



Sunk Costs

Sunk costs cannot be changed by any decision. They are not differential costs and should be ignored when making decisions.

Example: You bought an automobile that cost \$10,000 two years ago. The \$10,000 cost is sunk because whether you drive it, park it, trade it, or sell it, you cannot change the \$10,000 cost.



Quick Check ✓

Suppose that your car could be sold now for \$5,000. Is this a sunk cost?

- A. Yes, it is a sunk cost.
- B. No, it is not a sunk cost.



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